Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

pmsimstats team

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Abstract

Background. Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound, the patient's belief that the treatment will work, and natural-history drift in the underlying condition. Standard analysis models lump these into a single 'treatment effect' plus residual error, which conflates clinically distinct quantities and compromises both regulatory inference and predictive-biomarker validation.

Methods. We formalise a three-component decomposition of the within-participant outcome trajectory into a biological response component (BR), a placebo-belief response component (PB), and a time-variant natural-history component (TV), each modelled as a modified Gompertz function with participant-specific random effects. We characterise the trial-design contrasts (open-label, blinded discontinuation, crossover) that supply identifiability of each component, derive the linear-mixed approximations to the full nonlinear model that retain the pharmacological-component target estimand, and quantify the inferential cost of failing to decompose through worked examples and a Monte Carlo simulation study calibrated to the prazosin-PTSD reference parameters.

Results. The framework is developed and pre-registered for empirical evaluation; a 100-replicate pilot of the analysis machinery is reported, and a production simulation programme at 1,000 replicates per cell across $N \in \{35, 70, 100, 150\}$ is pre-registered for execution.

Conclusions. The three-component framework is the principled basis for predictive-biomarker validation in aggregated N-of-1 trials and clarifies the design contrasts that are mandatory versus optional. A pragmatic analysis specification consistent with the framework but tractable at trial-relevant sample sizes is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation.

1 Introduction

1.1 A claim about how N-of-1 trials should be analysed

[1] **This paper argues that treatment response in N-of-1 trials comprises three causally distinct components that must be analysed jointly rather than lumped.**

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each component is causally distinct, clinically interpretable, and identifiable from a well-designed N-of-1 trial.
- The standard single-indicator mixed-effects model conflates all three and renders biomarker validation unanswerable.

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This paper advances an opinionated methodological claim: treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components and should routinely be analysed as such, rather than estimated as a single lumped quantity. The three components are pharmacological action (BR, the biological response), placebo-belief response (PB, the patient's belief that they are receiving treatment), and time-variant natural history (TV, the trajectory that would have occurred without intervention). Each is causally distinct, each is independently of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. We shall argue that the standard practice of fitting a single mixed-effects model with one treatment indicator, prevalent across the published N-of-1 literature, conflates these three components, biases the interpretation of treatment effect, and renders predictive-biomarker validation unanswerable as a scientific question.

[2] **The contribution is the consolidation and extension of an established structural-decomposition tradition, not a novel decomposition.**

- The additive disease-progression, placebo, and drug-action decomposition is well established in pharmacometrics and longitudinal trial methodology.
- The present framework extends that tradition with an explicit, separately identified placebo-belief channel and brings it to the aggregated N-of-1 setting.
- The decomposition logic is not specific to N-of-1; it applies wherever a longitudinal treatment response is analysed.

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We make no claim of novelty for the decomposition itself. The structural decomposition of a longitudinal treatment response into separable disease-progression, placebo, and drug-action components is a well-established tradition in pharmacometrics [@holford2015; @gomeni2009; @chen2021], and the separation of drug from placebo response in longitudinal trajectories has been pursued both through growth-mixture models [@muthen2009] and through experimental decomposition of the placebo response itself [@kaptchuk2008components]. Each ingredient we use (mixed-effects models with trial-phase indicators, placebo-arm contrasts, regression-to-mean adjustment, growth-curve parameterisation) has likewise appeared across decades of clinical-trial methodology. Our contribution is therefore one of consolidation and extension rather than invention: we make the three-component structure explicit for the aggregated N-of-1 setting, add a separately modelled placebo-belief channel identified by design contrasts rather than by functional form alone, and argue that the decomposition should become routine wherever a longitudinal treatment response is analysed. We document why the framing has not been routine in N-of-1 practice (parsimony, identifiability concerns, sample-size constraints, methodological inertia) and propose an analysis that is tractable at trial-relevant sample sizes through a phase-augmented linear-mixed-effects approximation to the full decomposition.

1.2 Why this matters: three failure modes of the lumped analysis

[3] **The lumped one-component analysis has three documented failure modes that grow with unmodelled component heterogeneity.**

- The three failure modes are placebo-attribution bias, regression-to-the-mean, and biomarker-validation failure.
- Each failure mode is documented in the chronic-condition trial literature, not merely hypothetical.
- The failures are present regardless of sample size.

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We begin with the failure modes of the standard approach. A one-component analysis ('the treatment effect is what we observe minus baseline') has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodelled components. None of these is hypothetical: each has a documented presence in the chronic-condition trial literature.

[4] **Placebo-attribution bias is sample-size-invariant and documented across antidepressant, anxiety, and IBS literatures.**

- Lumped treatment-effect estimates overstate pharmacological efficacy in proportion to PB magnitude and variance.
- Kirsch et al. and Locher et al. demonstrated that antidepressant apparent efficacy is largely explained by placebo response and natural fluctuation.
- Placebo responsiveness has measurable biological correlates (e.g. COMT genotype), confirming it is a structured phenomenon, not noise.

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The first is the placebo-attribution failure. When the population contains both high and low placebo responders, the lumped treatment effect overestimates the pharmacological component. The bias scales with the magnitude and variance of PB in the study population, and it is invariant under sample-size increases: a larger one-component trial converges on the same biased estimate. The most consequential example in modern clinical pharmacology is antidepressant efficacy. @kirsch2008 analysed FDA-submitted data on four widely-prescribed antidepressants and showed that, except at the most severe baseline-severity stratum, mean drug-placebo differences fell below clinical-significance thresholds, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation; @locher2015 reproduced this pattern in late-life depression and explicitly invoked regression to the mean. @sugarman2014 generalised the finding to anxiety. Patient populations differ systematically in placebo responsiveness, and the difference has measurable biological correlates: COMT val158met genotype predicts placebo response in irritable bowel syndrome [@hall2012comt; @wang2022comt], indicating that 'placebo' is not a homogeneous nuisance but a structured biological phenomenon that interacts with the patient's genome and treatment history. A trial whose population happens to be placebo-responsive will report a larger 'drug effect' than one whose population is not, even when the underlying pharmacology is identical.

[5] **Regression-to-the-mean contributes several points of apparent improvement in early trial weeks and is absorbed without trace by the lumped analysis.**

- Enrolment at peak severity guarantees a natural-history regression component irrespective of pharmacology.
- Hrobjartsson et al. systematic reviews across 200+ trials showed much of conventional placebo response was indistinguishable from natural-history drift.
- The lumped analysis has no formal mechanism to recover the pharmacological effect once regression is absorbed.

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The second failure mode is regression-to-the-mean. When patients enrol at moments of clinical engagement that coincide with peak symptom severity, natural-history regression toward typical severity contributes several points of apparent improvement in the early weeks of any trial, without any pharmacological contribution at all. The systematic reviews of @hrobjartsson2001, @hrobjartsson2004, and @hrobjartsson2010, the last covering 202 trials across 60 clinical conditions, quantified this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The lumped analysis absorbs this regression into the treatment-effect estimate and the analyst has no formal mechanism to recover the true pharmacological effect.

[6] **Biomarker validation is invalidated when a biomarker correlates with PB or TV, rendering precision-medicine stratification wrong regardless of trial size.**

- A biomarker correlated with placebo responsiveness or natural-history trajectory is indistinguishable from a true pharmacological predictor in a lumped analysis.
- The PATH statement defers mechanistic attribution to domain knowledge; the BR-PB-TV decomposition supplies that substrate.
- Rigorously validated pharmacological biomarkers require the decomposition as a precondition, independent of sample size.

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The third failure mode is the biomarker-validation failure. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalised in the PATH statement [@kent2020path; @kent2020pathee]. A biomarker that correlates with placebo responsiveness or with a favourable natural-history trajectory will appear in a one-component analysis as a predictor of the lumped 'treatment effect', indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from such an analysis is wrong, and the trial's regulatory utility is degraded. The PATH framework explicitly defers the mechanism of treatment-effect heterogeneity to domain knowledge [@kent2018bmj; @rekkas2020scoping]; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that predictive-biomarker validation cannot be conducted rigorously in a lumped-analysis framework regardless of how large the trial is.

1.3 The current state of the N-of-1 literature

[7] **The N-of-1 literature has matured substantially but motivates three observations that the present paper addresses.**

- The design has evolved from individual-patient consultation to a framework supporting population-level inference.
- CONSORT reporting standards and quality audits document progressive improvement in analytic transparency.
- Three specific gaps in the literature motivate the decomposition framework proposed here.

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The N-of-1 trial in its modern randomised form originates with @guyatt1986, who defined the within-patient multiple-crossover RCT, demonstrated it on a single airflow-limitation case, and established a clinical service to deliver it at scale. The design inherits its identifying logic from two older traditions: the crossover trial of classical biostatistics, in which within-subject period contrasts estimate treatment effects under explicit carryover assumptions, and the single-case experimental designs of behavioural science, in which phase reversals (withdrawal and reinstatement of treatment) identify effects within one participant. The blinded-discontinuation contrast central to this paper is a direct descendant of both. The methodological step from the individual-patient trial to population-level inference was taken by @zucker1997 and @zucker2010, who showed that a series of single-patient trials can be combined through hierarchical meta-analysis to estimate both population treatment effects and individual responses, the statistical foundation on which the aggregated design rests. Over the last fifteen years the design has accordingly been reframed from an individual-patient decision tool into an aggregated, meta-analytic framework supporting population-level inference: the programmatic statement of @lillie2011, the methodological reviews of @duan2013, the historical-practice audit of @gabler2011, and the more recent syntheses of @schork2017nutrition, @schork2018rct, @schork2023precision, and @porcino2022 trace that trajectory. Reporting standards are codified in the CONSORT extension @vohra2015cent (the headline statement) and @shamseer2015cent (the explanation-and-elaboration companion); the reporting-quality audits of @li2016cent and its successors document slow but progressive improvement in analytic transparency. Three observations about this literature motivate the present paper.

[8] **No published N-of-1 trial employs a structural decomposi-**

tion of treatment response despite three decades of methodological literature.

- Audit data show the field has relied on paired t-tests, binary-indicator LME, and hierarchical Bayesian regressions.
- Recent methodological advances (Liao 2023, Blackston 2019, Carrozzo 2025) retain the lumped-response framing.
- The decomposition gap persists even as carryover and identifiability problems are increasingly recognised.

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The first observation is that the dominant analytic models are simple. The reporting-quality audit of @li2016cent documents that the published N-of-1 trials between 1985 and 2013 relied predominantly on paired t-tests, simple linear mixed-effects models with a binary treatment indicator, or hierarchical Bayesian regressions. On our own reading of that corpus and the methodological literature that followed it, no published N-of-1 trial has employed a structural decomposition of treatment response into pharmacological, expectancy, and natural-history components. The most recent methodological advances retain this character: @liao2023 develops Bayesian distributed-lag models with autocorrelated errors as a current carryover-aware analysis, but the model still treats the response as a single quantity to be estimated. @blackston2019 and @carrozzo2025 compare aggregated N-of-1 trials with parallel and crossover RCT alternatives by simulation and document carryover-induced type-I-error inflation as a key failure mode, but again at the level of the lumped treatment effect. None of these papers decompose the response.

[9] **Placebo-component separation is widely recognised but almost entirely unaddressed at the protocol level in N-of-1 methodology.**

- The Hendrickson hybrid blinded-discontinuation phase is among the few protocol mechanisms that supply PB-versus-BR identifiability.
- Placebo run-in, open-label-placebo, and mediator-analysis approaches have not been imported into N-of-1 designs.
- The gap between awareness of the problem and protocol-level solutions motivates the present framework.

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The second observation is that the placebo-component separation problem is widely known but largely unaddressed in N-of-1 specifically. The blinded discontinuation phase used in the Hendrickson hybrid design [@hendrickson2020] is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of PB versus BR. Other relevant approaches, including placebo run-in designs, open-label-placebo trials, and mediator-analysis applications, have been developed in parallel but rarely imported into N-of-1 methodology.

[10] **Kaptchuk's within-placebo decomposition is the closest structural analogue; no published paper applies the corresponding full-response decomposition to N-of-1 trials.**

- Kaptchuk et al. (2008) empirically decomposed the placebo response into observation, ritual, and relationship components.
- Open-label placebo trials confirm that clinically meaningful PB-only response is possible even when patients know they receive placebo.
- The BR-PB-TV framework is structurally analogous but operates at the level of the full treatment response.

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The third observation is that the closest precedent in the literature is Kaptchuk's three-component decomposition of the placebo response itself. In a randomised trial of acupuncture-style placebo for irritable bowel syndrome, @kaptchuk2008components decomposed the placebo response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single 'expectancy' lump. The follow-up open-label placebo trials of @kaptchuk2010olp, @lembo2021olp, and @nurko2022 in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk's decomposition but applied at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone. To the best of our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients, including mixed-effects modelling, blinded-phase contrasts, and trajectory parameterisation, are all in routine use.

[11] **The decomposition gap is real, the tools to fill it are mature, and the failure modes are large enough to make it a methodological priority.**

- The gap between available methodology and current practice is documented and not attributable to missing tools.
- Failure modes in chronic-condition pharmacology are consequen-

tial, not academic.
- The remainder of the paper develops, applies, and evaluates the decomposition framework.

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The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the lumped analysis are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

1.4 The neurobiological and epidemiological reality of the components

[12] **Each component of the decomposition reflects a real causal mechanism documented in independent literatures.**

- BR corresponds to pharmacological action; PB to neurobiologically real belief-driven mechanisms; TV to natural-history drift.
- The components are not statistical constructs but distinct causal pathways with separate literatures.
- The section reviews the evidence for each component before establishing the formal model.

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The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

[13] **BR is the pharmacological target of regulation; PB is neurobiologically real and phase-modulated, which is precisely the design contrast that makes the decomposition identifiable.**

- PB is mediated by endogenous opioid/dopamine release, HPA-axis regulation, and top-down attentional shifts.
- Wager et al. neurologic pain signature provides a clean fMRI dissociation between pharmacological and expectancy contributions.
- Phase-dependent belief variation (open-label vs. blinded) is the design contrast that supplies PB identifiability.

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The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signalling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research [@benedetti2014; @frisaldi2020; @carlino2014; @carlino2016benedetti], mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is instructive here: @wager2013nps developed an fMRI-based 'neurologic pain signature' (NPS) that tracks nociceptive input and is reduced by the opioid agonist remifentanil, and the individual-participant meta-analysis of @zunhammer2018nps found that placebo treatments produce moderate reductions in reported pain but only negligible reductions in the NPS, providing a neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across patients with biological correlates [@hall2012comt; @wang2022comt] and is modulated by trial phase: when belief varies, as it does between open-label and blinded-discontinuation conditions, the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

[14] ***TV* is the untreated counterfactual trajectory; its heterogeneity is the central object of precision-medicine inference and must be modelled, not absorbed into noise.**

- TV sign depends on disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive conditions.
- Selection-driven enrolment at peak severity guarantees positive TV in early trial weeks regardless of disease class.
- The PATH statement identifies treatment-effect heterogeneity modelling as mandatory for precision-medicine inference.

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The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive (recovery on intrinsic timescale); for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive conditions it trends negative; and for selection-driven enrolment processes it is positive in the early weeks of the trial regardless of underlying disease class, because of regression to the mean [@hrobjartsson2001; @hrobjartsson2010]. The PATH statement [@kent2020path; @kent2020pathee] highlights that this heterogeneity is itself the central object of precision-medicine inference: heterogeneity of treatment effect must be modelled, not absorbed into noise, if the precision-medicine question is to be answered.

1.5 What this paper contributes

[15] **The paper makes four contributions spanning formalisation, identifiability mapping, tractable methodology, and quantification of the inferential cost of lumping.**

- Sections 2–3 formalise the decomposition; section 4 maps it to design contrasts; section 5 develops tractable analysis methods.
- Sections 6–7 provide worked examples and a Monte Carlo simulation study; section 8 applies the framework to biomarker validation.
- A companion pedagogical document provides extended worked examples for first-time readers.

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We make four contributions, each addressed in a section below. We begin in section 2 with the formal model and notation. Section 3 develops each component in turn, with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear-mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the worked examples, and section 7 presents the simulation study. Section 8 discusses applications to predictive-biomarker validation. The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` provides an extended worked-example treatment for readers approaching the material for the first time.

[16] **The first contribution is a formal specification of the three-component decomposition as a sum of Gompertz trajectories with participant-specific random effects.**

- Each component follows a modified Gompertz function; the joint trajectory is their sum plus observation-level noise.
- Cross-component covariances are permitted, allowing participant-level heterogeneity to co-vary across components.
- The formal specification is the substrate on which identifiability and analysis-side arguments rest.

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Our first contribution is to formalise the three-component decomposition for the N-of-1 trial setting (sections 2 and 3). Each component is modelled as a modified Gompertz function with participant-specific random effects; the joint trajectory is the sum plus observation-level noise; cross-component covariances are permitted. The formal specification is the substrate the framework builds on.

[17] **The second contribution maps identifiability of each component onto specific trial-design contrasts and characterises what is lost without them.**

- The hybrid open-label-plus-blinded-discontinuation design supplies identifiability of all three components.
- Designs lacking specific phases lose identifiability of the corresponding component.
- The mapping clarifies which design features are mandatory versus optional for each inferential goal.

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Our second contribution is to map identifiability onto trial-design contrasts (section 4). The standard hybrid open-label-plus-blinded-discontinuation design, together with phase-by-allocation contrasts, supplies identifiability of all three components. We characterise which designs do and do not supply this identifiability and what is lost in each case.

[18] **The third contribution is a phase-augmented linear-mixed-effects model that captures the BR-versus-PB attribution check at modest parametric cost.**

- The full nonlinear DGP-matched fit is computationally fragile and degree-of-freedom-expensive at $N \in [30, 150]$.
- The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.
- This model is the recommended default for trial-relevant sample sizes.

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Our third contribution is to develop a tractable analysis methodology at trial-relevant sample sizes (section 5). The full nonlinear mixed-effects fit to mirror the DGP is computationally fragile at $N \in [30, 150]$ and consumes more degrees of freedom than trial-relevant sample sizes can spare. We propose a phase-augmented linear-mixed-effects model

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
```

```
        + bm:Dbc + bm:Dbc:phase
        + (1 | ptID),  correlation = corCAR1(...)
```

that captures the practically important part of the decomposition, specifically the BR-versus-PB attribution check via $Dbc : phase$ and $bm : Dbc : phase$, at modest parametric cost.

[19] **The fourth contribution quantifies the inferential cost of lumping through worked examples and a calibrated Monte Carlo simulation study.**

- Two participants with identical lumped responses but different (BR, PB, TV) compositions are indistinguishable under the one-component analysis.
- The simulation study quantifies bias, type I error, and power across analysis strategies and parameter regimes.
- The calibration uses prazosin-PTSD reference parameters from the Hendrickson et al. dataset.

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Our fourth contribution is to quantify the inferential cost of failing to decompose, through worked numerical examples (section 6) and a Monte Carlo simulation study (section 7) calibrated to the prazosin-PTSD reference parameters. The worked example shows how two participants with identical lumped responses but different underlying (BR, PB, TV) compositions appear identical to a one-component analysis and qualitatively different under the decomposition. The simulation study quantifies bias, type I error, and power across the analysis strategies and across the parameter regimes most relevant to chronic-condition trials.

[20] **The decomposition is not optional for predictive-biomarker validation: it is the formal mechanism by which the pharmacological precision-medicine question can be asked at all.**

- A biomarker predicting the lumped total is not, in general, a pharmacological predictor.
- The decomposition separates the biomarker-by-BR question from biomarker-by-PB and biomarker-by-TV confounding.
- The decomposition should be the default, not an optional sensitivity analysis, in trials with biomarker validation aims.

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The framework's central application is to predictive-biomarker validation (section 8). A biomarker that predicts a lumped 'treatment effect' is not, in general, a pharmacological predictor. The decomposition is the formal mechanism that lets the analyst ask the precision-medicine question (does the biomarker predict BR?) rather than the lumped surrogate question of whether the biomarker predicts the total response, including placebo and natural history. We argue that the decomposition should be the default, not an optional sensitivity analysis, in any N-of-1 trial whose primary aim is biomarker validation.

2 Notation and the formal model

[21] **This section fixes compact notation and states the formal model; a fuller worked-example treatment is in the companion pedagogical document.**

- Definitions are precise enough to support the identifiability and analysis-side arguments in later sections.
- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

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In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

[22] where Y_{it} is the observed symptom score, BL_i is participant i's baseline score, the three components BR, PB, TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and ε_{it} is observation-level noise. The sign convention is that an increase in any component reduces Y; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

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For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it},$$

[23] **Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.**

- BR_{it} accumulates with time-on-drug; PB_{it} with time-on-belief scaled by expectancy η(phase); TV_{it} with calendar time from trial entry.
- The distinct time indices are the structural source of identifiability from phase-contrast data.
- All components are zero at $t = 0$ by construction and grow over the trial duration.

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Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation. PB_{it} depends on time-on-belief, modulated by a phase-specific expectancy factor η(phase) that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

[24] rescaled so that $C(0) = 0$ and $C(t) \to m_C$ as $t \to \infty$, for $C \in \{BR, PB, TV\}$. The three Gompertz parameters (m_C, d_C, r_C) are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-person response variability that the framework is designed to characterise.

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Each component follows a modified Gompertz curve:

$$C(t) \;=\; m_C \,\exp(-d_C \exp(-r_C t)) \;-\; m_C \,\exp(-d_C),$$

[25] **The residual term has AR(1) structure and the three components are permitted to co-vary through cross-component covariances.**

- Within-participant residuals follow $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.
- Within-time and across-time cross-component covariances permit participant-level heterogeneity in one component to co-vary with the others.
- This structure accommodates biologically plausible correlations, such as high-BR patients also tending toward high-PB.

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Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient's belief that they are receiving treat
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of
η(phase)	Expectancy multiplier for PB: 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, between
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$.

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ε_{it} is a within-participant residual term with AR(1) serial-correlation structure $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances c_{cf1t} and c_{cfct} respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarised in Table 1.

3 The three components

[26] **This section presents each component with its biological motivation, Gompertz parameterisation, and identifiability conditions to establish the model as causally interpretable.**

- The three components are presented in turn: BR (biological response), PB (placebo-belief), TV (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.
- The section's purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

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In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

[27] **BR is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.**

- BR arises from chemical action on the body and is zero for inert pills regardless of patient expectation.
- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating BR from PB and TV is the central precision-medicine problem the framework addresses.

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BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, suggestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

[28] **PK/PD considerations produce a saturating BR trajectory with a slow initial rise, a peak-growth phase, and a sharp drop to zero on discontinuation.**

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

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The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully [@raskind2013prazosin]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

[29] **The three Gompertz parameters (m_{BR}, r_{BR}, d_{BR}) capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.**

- m_{BR} is the biological ceiling (maximum reduction at infinite exposure); r_{BR} the onset rate; d_{BR} the displacement shaping the early climb.
- Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42$/week, $d_{BR} \approx 5$.
- These are participant-specific random effects whose population distributions encode between-person response heterogeneity.

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The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR}t)) - m_{BR} \exp(-d_{BR})$ captures this profile compactly. The maximum m_{BR} is the biological ceiling, the largest reduction the drug can produce at infinite exposure. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the @hendrickson2020 reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

[30] **The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.**

- Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded growth.
- Exponential $BR = m(1 - e^{-rt})$ lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-parameter form avoids overfitting; sensitivity analysis is reported

separately in `07-gompertz-evaluation`.

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Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1 - \exp(-rt))$ is acceptable for fast-onset drugs but lacks the slow-start phase that characterises most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modelling. The sigmoid E_{max} (Hill) model [@holford2011backstory] describes the dependence of effect on concentration (a dose-response curve), whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`.

[31] **BR is identified by the blinded-discontinuation contrast, where drug removal collapses BR to zero while PB remains intact.**

- Within-participant on-drug versus off-drug contrasts are the primary identification mechanism.
- The blinded-discontinuation phase provides the cleanest contrast: BR drops, PB is preserved.
- This contrast does not require the analyst to model PB or TV explicitly to recover BR.

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The BR component is identified by within-participant on-drug versus off-drug contrasts, formalised in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR collapses to zero while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly.

3.2 Placebo-belief response (PB)

[32] **PB is the strict placebo response: improvement that persists when the active drug is silently replaced, provided the patient's belief is unchanged.**

- PB is belief-dependent, not drug-dependent; it would survive a silent switch to placebo under sustained belief.
- It is distinct from BR in causal origin: BR requires chemical action; PB requires only belief.
- The definition is strict and operational, not vague or colloquial.

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PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

[33] **PB reflects neurobiologically real mechanisms triggered by belief, not pharmacology; neuroimaging and genetic evidence confirm it is a structured biological phenomenon.**

- Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down attentional shifts are all documented mechanisms.
- Wager et al. neurologic pain signature dissociates pharmacological from expectancy contributions via fMRI.
- COMT genotype predicts placebo responsiveness, confirming biological structure rather than measurement artefact.

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It is worth being precise about what PB is and is not. PB is not 'fake' improvement, and it is not 'all in the patient's head' in any dismissive sense. The placebo response is mediated by well-characterised neurobiological pathways [@finniss2010; @benedetti2014; @frisaldi2020; @carlino2014; @carlino2016benedetti]. Belief-driven release of endogenous opioids and dopamine produces measurable analgesia and mood elevation; anticipatory regulation of the hypothalamic-pituitary-adrenal axis modulates cortisol and inflammatory cytokines; top-down attentional shifts change how symptoms are perceived and reported. Each of these mechanisms is a real biological response. What makes them 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology. The neuroimaging evidence is instructive: @wager2013nps developed an fMRI-based 'neurologic pain signature' that is reduced by remifentanil, and @zunhammer2018nps showed across pooled participant-level data that placebo treatments reduce reported pain substantially while leaving the signature essentially unchanged, dissociating the pharmacological and expectancy contributions to a clinical pain response. Genetic correlates of placebo responsiveness [@hall2012comt; @wang2022comt] further establish that placebo response is a structured biological phenomenon, not a measurement artefact.

[34] **PB is identifiable from BR only when belief and treatment allocation differ, as in the blinded-discontinuation phase.**

- The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether they remain on active drug.
- Belief drops to a partial level ($\eta \approx 0.5$) because the patient's expectation is hedged, not absent.
- Designs without a phase that decouples belief from treatment cannot separately identify PB from BR.

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For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

[35] **The $\eta = 0.5$ blinded-phase expectancy is a modelling assumption supported by literature evidence of 30–60% residual expectations, but requires sensitivity analysis.**

- Published placebo literature reports residual expectations of 30 to 60 percent of the open-label level across pain, depression, and PTSD trials.
- Open-label placebo trials confirm clinically meaningful PB-only responses even under explicit disclosure.
- Sensitivity analyses varying η are mandatory before clinical con-

clusions depend on this single parameter.

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PB also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB}\,\exp(-d_{PB}\exp(-r_{PB}t)) + (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and η between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The choice of $\eta = 0.5$ for the blinded phase is a modelling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that the residual response is substantial. Open-label placebo trials [@kaptchuk2010olp; @lembo2021olp; @nurko2022] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of η is not empirically pinned, sensitivity analyses varying it across its plausible range are mandatory before any conclusion is allowed to depend on this single number.

[36] **Kaptchuk et al. (2008) provide the structural precedent; the BR-PB-TV framework extends the same decomposition logic to the full treatment response.**

- Kaptchuk decomposed the placebo response itself into observation, ritual, and relationship components empirically.
- The present framework applies analogous decomposition logic at the level of drug, placebo, and natural history.
- Both frameworks reject the 'expectancy lump' model in favour of identifiable additive parts.

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The decomposition framework as proposed here has a structural analogue in the placebo-research literature. @kaptchuk2008components decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomised trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

[37] **PB variance scales with η, not just its mean; ignoring this heteroscedastic structure miscalibrates standard errors and downstream biomarker tests.**

- Phase-dependent variance follows $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$.
- Open-label phases have high between-patient PB variance; blinded phases have low variance.
- Ignoring heteroscedasticity deflates standard errors in open-label and inflates them in blinded phases, with consequences for CI coverage and biomarker test calibration.

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A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-suggestibility patients show large PB, low-suggestibility patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the patient-to-patient variance shrinks accordingly. The model therefore parameterises $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

[38] **Additivity is a modelling assumption, not biological fact; the section examines whether a $BR \times PB$ interaction term is warranted.**

- The formal model writes total response as a sum; this section asks whether the BR-PB interaction is zero.
- The question has inferential consequences for biomarker validation and for what the phase-augmented analysis captures.
- The treatment is preparatory for the generalised interaction model introduced later in this section.

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In this section we shall broach a question that the formal model leaves open: whether the three components contribute additively to symptom reduction, or whether there is a meaningful interaction between BR and PB that the additive specification misses.

[39] **The additive model may be wrong: published literature documents subadditivity between pharmacological and placebo-belief effects in some clinical contexts.**

- Subadditivity means the combined $BR + PB$ effect is less than the sum of each component measured separately.
- Evidence from Kaptchuk et al. and Benedetti et al. shows the phenomenon is context-dependent, not universal.
- Formal test: $\partial Y/\partial(BR \cdot PB) > 0$ in a saturated mean-structure model.

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The decomposition as written in section 2 treats the three components as additive contributors to symptom reduction:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; (BR_{it} + PB_{it} + TV_{it}) \;+\; \varepsilon_{it}.$$

The additivity is a modelling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. Additivity is also scale-dependent in the sense emphasised by @senn2004: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent $BR \times PB$ interaction can be partly an artefact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by @bingel2011, who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanil while negative expectation abolished it, so the two channels are not in general separable by simple addition. More specifically, the published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, less than the sum of the two components measured separately [@kaptchuk2008components; @benedetti2014]. The phenomenon is typically termed subadditivity: $\partial Y/\partial(BR \cdot PB) > 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

[40] **Three candidate mechanisms for subadditivity are shared neural substrate, scale ceiling effects, and belief-dynamics**

adaptation.

- Shared pain-modulation circuits produce saturating occupancy when both BR and PB activate them simultaneously.
- Scale ceiling effects prevent registration of additional symptom reduction once $BR + PB$ saturates the measurement range.
- Belief-dynamics adaptation: unexpectedly large BR can recalibrate patient expectation downward, attenuating ongoing PB.

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Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, ceiling effects on the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR may attenuate PB via expectation recalibration.

[41] where $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form, and $\gamma > 0$ encodes superadditivity, is the natural extension. The interaction coefficient γ is identifiable from the same trial-design contrasts that identify the marginal components: open-label phases supply $BR \cdot PB$ products at high η, blinded phases supply $BR \cdot PB$ products at attenuated η, and the γ coefficient is estimated from the difference between the two regimes once the marginal components are accounted for.

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Whichever mechanism dominates in a given clinical context, the subadditivity question is empirical and design-dependent. Within the BR-PB-TV framework, subadditivity manifests as a non-zero coefficient on a $BR \times PB$ interaction term in the conditional-mean specification. The additive form treats this interaction as zero by construction; a generalised form

$$Y_{it} = \mathrm{BL}_i - [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] + \varepsilon_{it},$$

[42] **Subadditivity biases biomarker predictive accuracy estimates and is partly absorbed by the phase-augmented analysis even without explicit γ estimation.**

- Under subadditivity, the biomarker-by-BR association is partly cancelled, inflating apparent predictive accuracy in the additive model.
- Phase designs attenuating PB (blinded, crossover) push the effective biomarker-treatment estimate closer to the true biomarker-by-BR value.
- The phase-augmented analysis in section 5 absorbs subadditivity as a side benefit.

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For trial design, two consequences follow. First, a biomarker that predicts BR in an additive-decomposition analysis may overstate or understate predictive accuracy if the true mechanism is subadditive: under subadditivity, the biomarker-driven increase in BR is partly cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-treatment association is smaller than the biomarker-by-BR association. Second, trial designs that attenuate PB in some phases (blinded discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by-BR association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

[43] **A forthcoming simulation study will quantify how bias in the biomarker-treatment interaction scales with γ; the preregistration predicts linear scaling.**

- Bias in $\widehat{\beta}_{bm:D}$ is expected to scale linearly with γ at small $|\gamma|$.
- Under subadditivity, bias sign is opposite to the true biomarker-by-BR effect; under superadditivity it is aligned.
- Empirical quantification is deferred to the production simulation programme in section 7.

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A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ is forthcoming as part of the simulation programme described in section 7. The pre-registration is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by-BR effect under subadditivity and aligned with it under superadditivity.

[44] **Non-zero $\hat{\gamma}$ does not establish true subadditive interaction; latent-class structure with class-correlated components produces the same marginal pattern.**

- If class membership correlates with both BR and PB strength, the marginal covariance of $BR \cdot PB$ mimics a direct interaction term.
- This confounding is invisible from the lumped-analysis perspective but is identifiable in a class-aware analysis.
- Apparent subadditivity can arise from latent population heterogeneity with no direct BR-PB interaction in the class-conditional DGP.

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A subtle but inferentially important point merits attention here: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not, on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modelling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in `analysis/report/03-latent-class-mixture-application/`), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the lumped-analysis perspective this looks exactly like a γ-weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct BR-PB interaction.

[45] **Distinguishing true subadditivity from latent-class confounding requires either a class-aware analysis or the phase contrasts the BR-PB-TV framework provides.**

- The two mechanisms produce qualitatively similar marginal effects but are inferentially distinct.
- $\hat{\gamma}$ should be reported alongside a class-aware sensitivity analysis, not as standalone evidence for mechanistic subadditivity.
- The latent-class paper's `pb_class_correlation` simulation axis

is the direct test; cross-paper synthesis is the cleanest disentanglement.

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The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

3.3 Time-variant natural-history component (TV)

[46] **TV is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.**

- TV is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from PB.

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TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

[47] **TV is a structured participant-specific signal, not noise; its distinguishing feature from PB is independence from treatment status and belief.**

- Unlike ε_{it}, TV is participant-level (slow trajectory) and requires explicit modelling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating TV.
- TV is modulated by neither treatment assignment nor patient belief, whereas PB is modulated by both.

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Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

3.3.1 When does natural history actually exist?

[48] **The shape of TV depends on disease class, trial timescale, and selection process; none of these can be assumed and all must be recovered empirically.**

- The assumption that subjects improve over time is true for some conditions and systematically wrong for others.
- Disease class, trial timescale, and enrolment selection jointly determine the sign and magnitude of TV.
- Recovering the empirical shape of TV is one of the decomposition's primary objectives.

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A common simplifying assumption in trial analysis is that subjects will improve over time, with or without treatment. This is true for some conditions and problematic for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

[49] **TV sign varies systematically by disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive, and structured for cyclical conditions.**

- Acute conditions (post-operative pain, acute back pain) resolve

intrinsically; substantial TV-improvement appears in the placebo arm.
- Progressive conditions (Alzheimer's, ALS) require explicit negative-trajectory modelling; apparent treatment effect is slowing of progression.
- Cyclical/triggered conditions (PTSD with trauma anniversaries, relapsing-remitting MS) require covariates beyond the Gompertz form.

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Acute conditions tend toward TV-positive: a sprained ankle, a post-operative pain syndrome, an episode of acute back pain all resolve on a timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV-improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV-zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV-negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

[50] **Selection-induced TV is real and rarely zero even for stable chronic conditions, because enrolment coincides with peak severity triggering help-seeking.**

- Regression to the mean alone contributes several points of apparent improvement in early trial weeks.
- Hrobjartsson et al. systematic reviews across 200+ trials show much of conventional placebo response is indistinguishable from natural-history drift.
- Kirsch (2008) and Locher (2015) demonstrate the clinical consequences in antidepressant and late-life depression trials.

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Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enrol in trials at moments of clinical engagement, which often coincide with recent worsening that triggered help-seeking. Regression to the mean alone produces several points of apparent TV-improvement in the early weeks of a trial, even in a population whose underlying condition is stable. The systematic-review evidence of @hrobjartsson2001, @hrobjartsson2004, and @hrobjartsson2010, the last covering 202 trials across 60 clinical conditions, quantifies this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The depression literature provides the most consequential modern example: @kirsch2008, analysing FDA-submitted antidepressant trials, showed that mean drug-placebo differences fell below clinical-significance thresholds except at the most severe baseline-severity stratum, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation; @locher2015 explicitly invoked regression to the mean to explain the severity-by-treatment interaction in late-life depression.

[51] **Trial participation itself induces TV-improvement through Hawthorne, structured-visit, and diary-completion effects, independent of pharmacology or placebo.**

- Hawthorne, structured-visit, and diary-completion effects each contribute symptom improvements attributable to participation, not treatment.
- These contributions are absorbed into TV in the decomposition, not into BR or PB.
- Ignoring these participation-induced effects inflates the apparent pharmacological and placebo components.

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Trial participation itself also induces TV: the Hawthorne effect, the structured-visit effect, and the diary-completion effect appear in the data as TV-improvement that is independent of any pharmacological or placebo response.

3.3.2 Can we assume no TV effect?

[52] **Assuming $TV = 0$ is rarely safe in chronic psychiatric or pain trials; the conservative default is to estimate TV from the data.**

- Zero-TV assumption is valid only when all four conditions hold: stable disease, short trial, non-severity-peaked recruitment, no ancillary care.
- For most chronic conditions at least one condition fails; the assumption should be tested, not asserted.
- Statistical efficiency lost by estimating TV is smaller than the

bias introduced by assuming it zero.

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But how safe is it to assume TV is zero? The short answer is: rarely safe, and never safe without empirical support. The longer answer is that TV may be assumed zero only if the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favour participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than assume it is zero.

[53] **The decomposition makes the zero-TV assumption testable and shows that lumping TV into residuals produces autocorrelated, heteroscedastic residuals that bias BR and PB standard errors.**

- $\hat{m}_{TV}$ with a variance estimate from the model provides an empirical test of the zero-TV assumption.
- Participant-level TV is structured and correlated with the on/off-drug schedule; it is not independent noise.
- Lumping TV into ε produces autocorrelated within-participant residuals and heteroscedastic cross-participant residuals.

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The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical support for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and lumping it into ε is worse than estimating it: the participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so lumping it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

3.3.3 Why a Gompertz, not a linear time covariate?

[54] **A linear `week` covariate is insufficient for TV because it**

enforces linearity, ignores participant heterogeneity, and cannot distinguish TV from BR without the explicit decomposition.

- A `week` covariate constrains each additional week to contribute a fixed natural-history change, misrepresenting curved or saturating disease trajectories.
- A single coefficient estimates a population-mean trajectory, ignoring participant-level heterogeneity in natural history.
- Without the decomposition and design contrasts, `week` absorbs both BR and TV, making them inseparable.

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A natural objection is that a model with `week` as a covariate already controls for time. But how adequate is such a control? Three reasons suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, BR and TV both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

4 Identifiability and trial-design contrasts

[55] **Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.**

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterises the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of BR, PB, and TV.

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The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterise the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

[56] **Three phase-by-allocation contrasts isolate PB, BR, and TV respectively; the LME model combines them but the inference is driven by design.**

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component.
- Blinded on-drug versus blinded placebo yields BR alone (since PB and TV match across arms within the blinded phase).
- Off-drug phases yield TV plus residual off-drug PB, with no BR contribution.

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Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component. The blinded on-drug versus blinded placebo contrast yields BR alone, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB, with no BR contribution and PB at the off-drug expectation level.

[57] **The LME model formally combines all three contrasts, but inferential precision is determined by the design, not model specification.**

- Each contrast targets a specific component; the model integrates all three into a coherent parameter set.
- Model specification cannot compensate for missing design contrasts; the identifying information must exist in the data.
- The analyst should evaluate any proposed trial design by verifying each component has at least one identifying contrast.

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Each contrast targets a different component, and the analyst combines them through the linear-mixed-effects model fitted to the full trajectory. The model is the formal mechanism, but the inference is driven by the design contrasts.

[58] **Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.**

- A pure open-label design cannot separate BR from PB because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ PB because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

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A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

[59] **Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.**

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.
- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for TV to separate from a decaying BR.
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

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Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR, the open-label-versus-blinded contrast for PB, and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR. Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the assumed η(phase) multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

5 Analysis-side methodology

[60] **Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.**

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

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In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

[61] **The first constraint is identifiability: forcing a three-component model on a design with insufficient contrasts produces unidentified parameters, not component estimates.**

- A pure open-label trial cannot distinguish BR from PB; both rise from zero together on treatment initiation.
- Fitting three components to a one-contrast design produces implausibly tight standard errors that misrepresent inferential precision.
- The analyst must audit whether the design supplies the necessary contrasts before specifying a component-matched model.

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The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

[62] **The second constraint is degrees of freedom: a component-matched analysis adds approximately a dozen non-linear parameters, inflating moderation estimand variance at $N \in [30, 150]$.**

- Gompertz BR (3), PB (3 + expectancy multiplier), and TV (3) each require participant-specific random effects on the maxima.
- At trial-relevant N, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts.
- Additional parametric structure typically inflates moderation estimand variance more than it reduces bias.

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The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation programme) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\widehat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

[63] **The third constraint is convergence fragility: component-matched nonlinear mixed models converge below 90% at trial-relevant N, contaminating aggregated analyses.**

- Simultaneous Gompertz BR, PB, and TV fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

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The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant N, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

[64] **The fourth constraint is estimand robustness: the `bm:Dbc` marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.**

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal `bm:Dbc` interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

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The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

[65] **The inferential target is stated as an estimand: the population biomarker-by-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.**

- The $bm{:}D_{bc}$ estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modelled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

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It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework [@iche9r1], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-by-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarised through the continuous indicator D_{bc}, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modelled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR}

is a more ambitious estimand: it is the biomarker-by-BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasised for individual treatment effects [@senn2004; @senn2018]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 A small extension that usually pays off: phase indicators

[66] **The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.**

- `phase` absorbs systematic differences in PB expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of PB-mediation.
- `bm:Dbc:phase` detects whether the biomarker-by-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by-PB.

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The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),  correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

[67] **The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation**

diagnostic without requiring the Gompertz parametric form.

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate BR and PB components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

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This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form.

5.3 When to fit the full decomposition

[68] **The full decomposition merits the effort only when N is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.**

- Feasibility conditions: several-hundred N, dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.
- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

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A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

[69] **The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterise how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.**

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterisation tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

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The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

[70] **The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.**

- The continuous indicator D_{bc} decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

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A word on the carryover parameterisation is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al. [@liao2023] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza [@daza2018] formalises an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and Gärtner et al. [@gartner2023] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation programme of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

6 Worked example: information loss without decomposition

[71] **The worked example uses two participants with identical lumped responses but opposite BR-PB profiles to show that the one-component analysis cannot distinguish them.**

- Participant A has high PB ($m_{PB} = 6$) and moderate BR ($m_{BR} = 4$); Participant B has low PB ($m_{PB} = 1$) and high BR ($m_{BR} = 8$).
- Both participants reach approximately 10–11 points of total improvement by week 8, making them indistinguishable to a one-component analysis.
- The blinded-discontinuation phase creates the diagnostic contrast that exposes the underlying difference.

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To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m^A_{BR} = 4$, $m^A_{PB} = 6$, $m^A_{TV} = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m^B_{BR} = 8$, $m^B_{PB} = 1$, $m^B_{TV} = 1$.

[72] **Open-label totals are clinically indistinguishable; the one-**

component analysis estimates a population drug effect of approximately 10 points with no differentiation between A and B.

- Both participants show approximately 10–11 points of improvement and present as clinical successes.
- A t-test or simple LME on open-label change cannot detect the heterogeneity in BR versus PB composition.
- The population drug-effect estimate conflates both pharmacological and placebo contributions.

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In the open-label phase (weeks 1 to 8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

| | BR | $PB(\eta = 1)$ | TV | Total improvement | |—|—|—|—|—| | Participant A | 4.0 | 6.0 | 1.0 | **11.0** | | Participant B | 8.0 | 1.0 | 1.0 | **10.0** |

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

[73] **The blinded-discontinuation phase reveals the BR-PB heterogeneity: Participant A retains 4 points while B collapses to 1.5, exposing the diagnostic contrast.**

- Participant A's high PB at $\eta = 0.5$ (3 points) sustains improvement even without active drug.
- Participant B's low PB (0.5 points) and absent BR produce near-complete return to baseline.
- The 4.0 versus 1.5 contrast is exactly what the decomposition uses to separate BR from PB.

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In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that BR has effectively cleared by the end of week 12; the more realistic case of a slowly decaying BR, modelled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

| | BR | $PB(\eta = 0.5)$ | TV | Total improvement | |—|—|—|—|—| | Participant A | 0.0 | 3.0 | 1.0 | **4.0** | | Participant B | 0.0 | 0.5 | 1.0 | **1.5** |

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds TV at its week-8 value; in reality TV continues to accumulate over weeks 9 to 12, which shifts both totals by a common amount and leaves the BR-versus-PB contrast on which the example turns unchanged.

[74] **The one-component model estimates drug at 7.75 points with no principled decomposition, obscuring that A's true BR is 4 and B's is 8.**

- The 7.75 point estimate is a conflated average of true BR values (4.0 and 8.0) plus the placebo wash-out.
- The model cannot determine whether A and B differ, nor estimate the pharmacological drug effect separately.
- Placebo and natural-history effects are not identifiable from this contrast alone.

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A one-component fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B). - Blinded-placebo means: 4.0 (A), 1.5 (B). - Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75. - Estimated placebo effect: not identifiable from this contrast. - Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B; it estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no principled way to decompose further.

[75] **The full decomposition recovers all six component estimates by exploiting phase differential expectancy and the blinded-placebo contrast.**

- The LME returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$ at the population point estimate.

- Participant-level estimates have meaningful uncertainty at trial-relevant N; the simulation study characterises this.
- The phase differential ($\eta = 1$ vs. $\eta = 0.5$) is the identification mechanism for the PB parameter.

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A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$, $\hat{m}^A_{TV} = \hat{m}^B_{TV} = 1.0$ (at the population point estimate; participant-level estimates have meaningful uncertainty at N in the trial-relevant range, which the simulation study below characterises).

[76] **The one-component model produces one biased average; the three-component model produces six clinically actionable quantities, making biomarker validation possible.**

- The lumped estimate of 7.75 points is a biased average of $BR_A = 4$ and $BR_B = 8$ confounded with placebo wash-out.
- The decomposition supplies individual BR, PB, and TV estimates enabling per-participant clinical decisions.
- Predictive-biomarker validation requires the decomposition because biomarker-by-BR is the target estimand.

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The information loss from lumping the components is not abstract. We may compare the inferences directly:

| Question | One-component answer | Three-component answer | |—|—|—| | Does the drug work for A? | 7.75 points (lumped). | $BR_A = 4.0$, distinct from her larger placebo response. | | Does the drug work for B? | 7.75 points (lumped). | $BR_B = 8.0$, twice the magnitude of A. | | Are A and B different responders? | No discernible difference. | B has 2x A's BR but 1/6 of A's PB. | | Is a biomarker predicting BR useful here? | Cannot answer. | Yes if the biomarker correlates with $\hat{m}_{BR}$. | | Should A be continued long-term? | Cannot say. | Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay. | | Population mean drug effect? | 7.75 points (biased). | $6.0 = (4.0 + 8.0)/2$ (correct mean of BR). |

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

[77] **A population-level example shows that cohort differences in open-label totals driven entirely by PB produce apparent non-replicability that the decomposition resolves.**

- Cohort 1 (naive, $m_{PB} = 7$) reports 13-point improvements; Cohort 2 (sceptical, $m_{PB} = 2$) reports 8 points; true $m_{BR} = 5$ in both.
- A one-component analysis would falsely conclude the drug works less well in Cohort 2 and trigger searches for confounders.
- The decomposition returns $\hat{m}_{BR} = 5$ in both cohorts, attributing the difference correctly to PB heterogeneity.

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A second example at the population level illustrates the same problem. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort 1 (early trial, naive patients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2 (later trial, patients who have had multiple prior failed treatments and are sceptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$. By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug works less well in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in PB, specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. Unfortunately, this is the kind of apparent non-replicability that pharmacology has spent the last decade struggling to navigate; the decomposition is one of the formal tools that makes it tractable.

7 Simulation study

[78] **The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.**

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registrat`
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

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The simulation study reported and pre-registered in this section is structured under the @morris2019 ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

7.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

[79] **The pilot is a structural check of the analysis pipeline, not a substantive empirical result.**

- The production simulation in section 7.2 constitutes the empirical core of the manuscript.
- The pilot confirms only that data generation, model fitting, and summary output run without error.
- No bias or power claims are derived from the 100-replicate pilot.

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The production simulation programme described in the following section is the empirical core of this manuscript and has not yet been executed. The pilot reported here confirms only that the analysis machinery runs end-to-end.

[80] **The pilot executed a 6-cell, 100-replicate run with perfect convergence, confirming the pipeline is operational.**

- The grid crossed three *pb_strength* levels against two analysis strategies.
- Convergence was 1.0 across all six cells under 8-core `mclapply`.
- Output is checkpointed at `analysis/data/quick-sim/`; no inferential claims are drawn.

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A 100-replicate-per-cell pilot was run at the reference Hybrid OL+BDC design, $N = 35$, on the 6-cell slice $pb\_strength \in \{0, 6.5, 13\} \times$ analysis $\in$ {one-component, phase-augmented}, using 8-core `parallel::mclapply`. The convergence rate was 1.0 across all six cells. Per-replicate output is checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds` and the

Morris-format cell-level summary at `analysis/data/quick-sim/06-decomp-summary.txt`. The exploratory summaries are diagnostic only. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across $pb\_strength \in \{0, 6.5, 13\}$, while the phase-augmented analysis showed 0.65, 0.65, and 0.61; the mean estimated $\hat{\beta}_{bm:D_{bc}}$ was essentially flat within each analysis (about -2.25 for the one-component and -2.21 for the phase-augmented fit). At 100 replicates the Monte Carlo standard error on a power estimate near 0.7 is about 0.046, so these figures cannot by themselves separate the two analyses; their interpretation, including why the phase-augmented analysis loses its bias-detection contrast at this N, is taken up in the Discussion. No confirmatory bias or power claim is drawn from the pilot, and all substantive claims are deferred to the production programme below.

[81] **Two pilot findings revise the production specification: rank-deficiency forces a formula change, and bias must be read from coefficient means, not rejection rates.**

- At $N = 35$, rank-deficiency dropped the D_{bc}:phase term, eliminating the key biomarker-by-treatment-by-phase contrast.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not in the Wald-test rejection rate.
- The production programme addresses both by using larger N, a fuller design, and coefficient-mean reporting.

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Two design-level observations from the pilot inform the production specification. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the D_{bc}:phase interaction term, removing the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect. Second, the predicted bias of the one-component analysis arises in the regression coefficient $\hat{\beta}_{bm:D_{bc}}$, not in the indicator-level rejection rate, so power on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The production programme below addresses both observations through larger N, a richer trial design that admits the full phase-augmented formula, and reporting of the coefficient mean against the implied true value rather than the rejection rate alone.

7.2 Production-simulation specification

[82] **The production simulation is organised around four deliverables under the ADEMP framework.**

- Driver scripts are located at `analysis/scripts/component-decomposition/`.

- The ADEMP pre-registration is committed prior to production runs.
- The four deliverables address bias, identifiability, type I/power, and participant-level recovery.

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The production programme follows the @morris2019 ADEMP framework. The driver scripts will live at `analysis/scripts/component-decomposition/`. We shall describe the four deliverables in turn.

[83] **The first deliverable quantifies how much each analysis strategy biases the biomarker-treatment interaction estimate as placebo strength increases.**

- The estimand is mean $\hat{\beta}_{bm:D_{bc}}$ with Monte Carlo SE across 1,000 replicates per cell.
- Sample sizes $N \in \{35, 70, 100, 150\}$ are crossed with a *pb_strength* grid.
- The headline finding is the bias-versus-power trade-off invisible to the 100-replicate pilot.

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The first deliverable is the bias of the biomarker-treatment interaction estimate: mean and Monte Carlo standard error of $\hat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses, across sample sizes $N \in \{35, 70, 100, 150\}$ and a grid of *pb_strength* values, at $n_{\text{reps}} = 1{,}000$ per cell. The headline measure is the bias-versus-power trade-off that the pilot was unable to detect at $N = 35$.

[84] **The second deliverable assesses whether the full-decomposition analysis is identifiable at trial-relevant sample sizes.**

- Identifiability is measured via convergence rates of the nonlinear-mixed-effects fit.
- Variance-inflation factors on the Gompertz parameters diagnose collinearity across components.
- Sensitivity of the BR estimate to η(phase) multipliers is reported explicitly.

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The second deliverable is a set of identifiability metrics for the full-decomposition analysis: convergence rates of the nonlinear-mixed-effects fit, variance-inflation factors on the component-specific

Gompertz parameters, and the sensitivity of the BR estimate to the chosen η(phase) multipliers.

[85] **The third deliverable characterises type I error and power across analysis strategies under a full sensitivity grid.**

- The DGP is the three-component reference; sensitivity axes are η mismatch, TV magnitude, and dropout.
- Type I error cells use 5,000 replicates to achieve MCSE 0.003.
- All three analysis strategies (one-component, phase-augmented, full-decomposition) are compared.

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The third deliverable is type I error and power for the biomarker-treatment interaction test under each analysis strategy, with the DGP held at the three-component reference and varied across a sensitivity grid (analyst-versus-truth η mismatch, TV-component magnitude, and dropout). Type I cells are run at $n_{\text{reps}} = 5{,}000$ for MCSE 0.003.

[86] **The fourth deliverable illustrates which analysis strategies preserve participant-level biomarker stratification.**

- Participant-level interaction signals are mapped as a function of (m_{BR}, m_{PB}, m_{TV}).
- The comparison reveals whether each strategy retains the predictive-biomarker signal.
- The lumped-total regression serves as the baseline that fails to stratify correctly.

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The fourth deliverable is the participant-level interaction signal recovered by each analysis, as a function of the underlying participant-level (m_{BR}, m_{PB}, m_{TV}) values, illustrating which analysis strategies preserve the predictive-biomarker stratification that the lumped-total regression does not.

[87] **Production runs use the pmsimstats-ng tidyverse collection with checkpointed outputs and a committed pre-registration.**

- The `tidyverse` implementation collection drives the simulation grid.
- Cell-level summaries are checkpointed under `analysis/data/`; per-replicate files under `output/`.
- The ADEMP pre-registration is committed at `00-ademp-pre-registration.md` before any production run.

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The simulation will use the existing pmsimstats-ng `tidyverse` implementation collection. Cell-level summaries will be checkpointed under `analysis/data/`, with per-replicate output under `analysis/data/quick-sim/06-decomp-replicates.rds` (pilot) and `analysis/scripts/component-decomposition/output/` (production). The ADEMP pre-registration is committed at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md` prior to the production runs.

8 Application to predictive-biomarker validation

[88] **The precision-medicine question – whether a biomarker predicts pharmacological response – cannot be answered without decomposing total response into its three components.**

- The question is whether the biomarker predicts the BR component, not the lumped total.
- The PATH statement formalises this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the formal mechanism for separating the pharmacological signal.

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The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalised in the PATH statement [@kent2020path], and it depends critically on the BR-PB-TV decomposition.

[89] **A lumped-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.**

- $\hat{\beta}_{bm}$ from the lumped regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If PB covaries with the biomarker distribution, flagged 'high responders' include high-PB responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

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A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

[90] **A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.**

- The sole criterion for pharmacological prediction is a meaningfully non-zero BR-component regression coefficient.
- A biomarker correlating only with PB selects placebo responders, not drug responders.
- A biomarker correlating only with TV is prognostic but not predictive in the pharmacological sense.

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The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

[91] **The decomposition is not optional for the precision-medicine question: it is the formal mechanism by which the question is posed at all.**

- Without separating BR from PB and TV, biomarker validation conflates pharmacological and non-pharmacological prediction.
- The phase-augmented linear-mixed analysis provides a small-N proxy via the `bm:Dbc:phase` interaction.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

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The decomposition is therefore not optional for the precision-medicine question. It is the formal mechanism by which the question is asked at all. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of this machinery at small N (the `bm:Dbc:phase` interaction tests whether the biomarker-by-treatment effect attenuates under blinding, which is the practical proxy for biomarker-by-PB contamination); the full decomposition recovers the component-specific regressions explicitly when N permits.

9 Discussion

[92] **The three-component framework reframes treatment response as a structured sum of causally distinct contributions, each carrying its own scientific and clinical interpretation.**

- The framework specifies which trial-design contrasts identify which components.
- The phase-augmented linear-mixed analysis implements the framework at trial-relevant sample sizes.
- The correct target estimand for biomarker validation is the regression of BR on the biomarker, not the lumped total.

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The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. The framework makes the conditions on trial design explicit (which contrasts identify which components), supplies an analysis-side methodology that accommodates trial-relevant sample sizes (the phase-augmented linear-mixed analysis), and clarifies the target estimand for biomarker validation (the regression of BR on the biomarker, not the lumped total).

[93] **Four open questions bound the framework's current scope: the simulation results are pending, the η multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.**

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The η(phase) expectancy-modulation parameters are modelling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic TV trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

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Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η(phase) multipliers used to parameterise the expectancy modulation of PB are themselves modelling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that @senn2018 and @senn2004 raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

[94] **Two companion documents extend the manuscript: a**

pedagogical narrative and a Gompertz-family sensitivity evaluation.

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

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The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

[95] **The decomposition is not specific to N-of-1; the same three-component structure and identifiability logic apply to any longitudinal design with belief-varying and exposure-varying contrasts.**

- The claim that a treatment response is the sum of pharmacological, belief-driven, and natural-history components is a statement about the data-generating process of any repeated-measures study.
- Parallel-group trials with repeated measures, crossover trials, and longitudinal observational cohorts all admit the required contrasts.
- The aggregated N-of-1 case is the most demanding instance because the components must be recovered from a single participant's trajectory.

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A further point concerns the scope of the decomposition. Although we have developed it for the aggregated N-of-1 design, nothing in the three-component structure is specific to N-of-1 trials. The claim that an observed longitudinal treatment response is the sum of a pharmacological component, a belief-driven component, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and the identifiability logic of section 4 transfers directly: wherever the design supplies a contrast in which belief varies independently of pharmacology (a blinded discontinuation, a placebo arm, an open-hidden administration) and a contrast in which drug exposure varies (on-drug versus off-drug, a crossover period), the same three components are separately identified. Parallel-group randomised trials with repeated measures, crossover trials, and longitudinal observational cohorts are all candidate settings, and the pharmacometric disease-progression models from which the present framework descends were themselves developed at the group level [@holford2015; @gomeni2009]. We therefore advocate the decomposition not as an N-of-1 technique but as a default lens for any longitudinal design in which placebo and natural-history contributions are non-negligible, with the aggregated N-of-1 case treated here as the most demanding instance because it must recover the components from a single participant's trajectory.

9.1 Interpretation of the pilot diagnostics

[96] **The pilot is a 6-cell, 100-replicate run designed to probe one-component bias under a three-component DGP, with MCSE approximately 0.046 on power estimates near 0.7.**

- The design crossed three *pb_strength* levels against one-component and phase-augmented analysis strategies.
- Per-replicate and Morris summary outputs are checkpointed at `analysis/data/quick-sim/`.
- At 100 replicates the Monte Carlo uncertainty is too large to draw reliable power comparisons.

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The pilot of section 7.1 was run specifically to probe the predicted bias of the one-component analysis under a three-component DGP; its design, convergence, checkpointed outputs, and Monte Carlo precision are reported there. We interpret its diagnostics here.

[97] **The pilot inverted the framework's power prediction, but the inversion is explained by rank-deficiency and a misidentified summary statistic rather than a failure of the theory.**

- At $N = 35$ rank-deficiency dropped the `Dbc:phase` term, so the phase-augmented analysis paid fixed-effect cost without gaining the bias-detection contrast.
- Predicted bias appears in the coefficient mean $\hat{\beta}_{bm:D_{bc}}$, which is flat across *pb_strength* at this resolution, not in the Wald rejection rate.
- Coefficient means (approximately -2.25 vs -2.21) cannot be separated from sampling noise with only 100 replicates at $N = 35$.

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The inverted finding reported in section 7.1, in which the phase-augmented analysis was *less* powerful than the one-component analysis across the entire *pb_strength* range, has two explanations. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the `Dbc:phase` interaction term, which removes the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect; without that term the phase-augmented analysis pays the cost of additional fixed effects without buying the bias-detection mechanism the framework promises. Second, the predicted bias of the one-component analysis arises in the regression *coefficient*, not in the indicator-level rejection rate. The mean estimated $\widehat{\beta}_{bm:D_{bc}}$ is essentially flat across *pb_strength* within each analysis (roughly -2.25 for the one-component analysis and -2.21 for the phase-augmented analysis), so the rejection rate on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias; the bias should be read off the coefficient mean itself, which the pilot's 100-replicate resolution does not separate from sampling noise at the $N = 35$ scale.

[98] **Production-rigor evaluation requires 1,000 replicates per cell, larger N, and recovery of the `Dbc:phase` contrast; the pilot is a structural, not inferential, check.**

- The canonical specification needs $N \in \{70, 100\}$ and a trial design that accommodates the full phase-augmented formula.
- The pilot confirms end-to-end pipeline operation on the three-component DGP but does not test the bias-detection claim.
- The section 7 production simulation remains the empirical core of the manuscript.

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A canonical run at 1,000 replicates per cell, larger N (70 and 100), and recovery of the `Dbc:phase` contrast (via a phase-stratified two-stage analysis or a richer trial design that admits the full formula without rank-deficiency) is required before the framework's bias-versus-power trade-offs can be quantified at production rigor. The pilot as configured is a structural check confirming that the analysis machinery runs end-to-end on the three-component DGP, not a substantive test of the manuscript's bias-detection claim. The section 7 simulation programme remains the empirical core of this manuscript.

10 Conclusions

[99] **Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and conflating them produces bias that scales with placebo and natural-history variance.**

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The bias from conflation grows with PB and TV variance in the study population.

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In conclusion, three points are to be emphasised. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation; conflating them into a single 'drug effect' produces estimates that are biased in ways that grow with the placebo and natural-history variance in the study population.

[100] **Decomposing response is identifiability-conditional: the trial design must supply phase contrasts, and the analyst must exploit them, but the phase-augmented linear-mixed analysis does this efficiently at small N.**

- Unique component identification requires phase contrasts built into the trial design.
- The phase-augmented linear-mixed analysis captures the decomposition without the parametric cost of a full nonlinear-mixed-effects fit.
- The full component-matched analysis is reserved for larger-N and stronger-prior settings where its inferential cost is justified.

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Second, decomposing the response is identifiability-conditional, not optional: the trial design must include phase contrasts that uniquely identify each component, and the analyst must use a methodology that exploits those contrasts. At trial-relevant sample sizes, a phase-augmented linear-mixed analysis captures the practically important part of the decomposition without paying the parametric cost of a full nonlinear-mixed-effects fit. The full component-matched analysis is reserved for the larger-N and stronger-prior settings where it is feasible and worth its inferential cost.

[101] **For predictive-biomarker validation the decomposition is not optional: it is the formal mechanism by which the precision-medicine question is posed.**

- A biomarker that predicts the lumped total is not, in general, a pharmacological predictor.
- Regulatory and prescribing decisions motivated by the trial require separating BR from PB and TV.
- The production simulation programme in section 7 is the planned empirical evaluation of these claims.

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Third, for predictive-biomarker validation the decomposition is not an optional refinement: it is the formal mechanism by which the precision-medicine question can be asked at all. A biomarker that predicts the lumped total is not, in general, a pharmacological predictor, and the regulatory and prescribing decisions that motivate the trial cannot be made without separating BR from PB and TV. We hope to return to the empirical evaluation of these claims in the production simulation programme described in section 7.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

[102] **The manuscript is rendered inside a Docker container with locked package versions; the full software environment is pinned by `Dockerfile` and `renv.lock`.**

- R 4.4.x in the zzcollab Docker container provides the execution environment.
- Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- The manuscript build uses `rmarkdown` and `bookdown`.

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[103] **Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds derived by index addition.**

- `set.seed(20260507)` is called at the top of each replicate loop.
- `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- Per-replicate seeds are `master_seed + rep_idx`.

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[104] **All simulation outputs, driver scripts, pre-registration, and manuscript source files are versioned at documented paths within the repository.**

- Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.

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Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates`
Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`.
Driver scripts are under `analysis/scripts/component-decomposition/`.
The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-reg`
The manuscript source, paper-specific bibliography (`references.bib`),
and SIM CSL file are at `analysis/report/06-component-decomposition/`.

15 References

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Source: `~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/06-component-decomposition/r`